

Central extracorporeal membrane oxygenation for refractory pediatric septic shock

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Objective: To demonstrate positive outcome, to achieve higher flow rates, and to reverse shock more quickly by implementing central extracorporeal membrane oxygenation (ECMO) in children with refractory septic shock. Children hospitalized with sepsis have significant mortality rates. The development of shock is the most important risk factor for death. For children with septic shock refractory to all other forms of therapy, ECMO has been recommended but estimated survival is <50% and the best method of applying the technology is unknown. In recent years, our institutional practice has been to cannulate children with refractory septic shock directly through the chest (central, atriaortic ECMO) to achieve higher blood flow rates.

Design: Retrospective case series.

Setting: Intensive care unit of a tertiary referral pediatric hospital.

Patients: Twenty-three children with refractory septic shock who received central ECMO primarily as circulatory support.

Interventions: Central ECMO.

Measurements and Main Results: The primary outcome measure was survival to hospital discharge. Pre-ECMO circulatory and

ventilatory parameters, infecting organism, duration and complications of ECMO and length of hospital stay were also collected. Twenty-three patients (median: age, 6 yrs; weight, 20 kg) over a 9-yr period were included. All patients had microbiological evidence of infection, and meningococemia was the most common diagnosis. Twenty-two (96%) patients had failure of at least three organ systems, and all patients received at least two inotropes with a mean inotrope score of 82.2 (sd, 115.6). Eight (35%) patients suffered cardiac arrest and required external cardiac massage before ECMO. Eighteen (78%) patients survived to be decannulated off ECMO, and 17 (74%) children survived to hospital discharge. Higher pre-ECMO arterial lactate levels were associated with increased mortality (11.7 mmol/L in nonsurvivors vs. 6.0 mmol/L in survivors, $p = .007$).

Conclusions: Central ECMO seems to be associated with better survival than conventional ECMO and should be considered by clinicians as a viable strategy in children with refractory septic shock. (Pediatr Crit Care Med 2011; 12:133–136)

KEY WORDS: sepsis; septic shock; pediatric; extracorporeal membrane oxygenation; cardiac arrest

Sepsis is one of the leading causes of death in children worldwide (1, 2). In poorer countries, many deaths may be attributable to malnutrition and inadequate health care, but mortality rates among children hospitalized with sepsis

in developed countries may still be as high as 10.3%, despite greater resource provision (3). In both developing and industrialized countries, the development of shock and multiorgan dysfunction syndrome is the most important determinant of mortality (2, 4). The American College of Critical Care Medicine has published guidelines to assist clinicians caring for children with septic shock (5). For shock unresponsive to all other therapies, they suggest that extracorporeal membrane oxygenation (ECMO) be considered. Standard venoarterial ECMO, with cannulas being placed in the blood vessels of the neck or groin, can be used to provide short-term, complete cardiac and respiratory support for children with sepsis (6–8), but the expected survival rate is <50% (5).

For the last 9 yrs at our institution, we have preferentially cannulated children with refractory septic shock through the chest with the aim of achieving higher flow rates and reversing shock more quickly. This involves performing a ster-

notomy and cannulating the right atrium and aorta (central ECMO), as opposed to the usual method of cannulating the femoral or jugular vein and the femoral or carotid artery (peripheral ECMO). Preliminary data showed that the central approach might be associated with superior outcomes (7). We now present a full review of all children placed on central ECMO for refractory septic shock at our institution. As septic shock in neonates is a different entity to septic shock in older children and is associated with better survival (5, 9, 10), we excluded neonatal patients from the study.

MATERIALS AND METHODS

The ECMO database of the pediatric intensive care unit, The Royal Children's Hospital, Melbourne, Australia, was used to identify all children who had received ECMO for septic shock from September 2000 until August 2009. All patient details are entered prospectively into the database, which is approved by the Institutional Review Board. Patients were included in the study if: they were placed on

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Table 1. Microbiology

Site	Organism	Number of Patients
Blood culture	<i>Neisseria meningitidis</i>	5
	<i>Streptococcus</i> species	3
	<i>Staphylococcus aureus</i>	2
	Coagulase-negative <i>Staphylococcus</i>	1
	<i>Burkholderia cepacia</i>	1
Blood polymerase chain reaction	<i>Klebsiella pneumoniae</i>	1
	<i>Neisseria meningitidis</i>	3
	<i>Staphylococcus aureus</i>	1
	<i>Aspergillus</i> species	1
Bronchoalveolar lavage culture	Influenza type B	1
Nasopharyngeal aspirate immunofluorescence		
Pleural fluid culture	<i>Streptococcus</i> species	2
Stool culture	<i>Salmonella</i> species	1
	<i>Campylobacter jejuni</i>	1

venoarterial ECMO for circulatory collapse despite intubation and mechanical ventilation, fluid therapy, and inotropes; had a clinical diagnosis of septic shock; had a positive bacterial culture, polymerase chain reaction, or viral immunofluorescence specimen taken before ECMO; and were aged >30 days and <18 yrs. The 2005 Pediatric Sepsis Consensus Conference criteria were used to define organ failure (11). Patients who were placed on ECMO primarily for respiratory failure were excluded. The primary outcome measure was survival to hospital discharge.

Our institutional practice for managing patients in septic shock has been described elsewhere (7). Goal-directed therapy is employed, as described in the American College of Critical Care Medicine guidelines (5). Clinical assessment, central venous oxygen saturation, lactate measurements, and echocardiography are used to guide therapy. If a patient has recalcitrant hypotension or organ dysfunction despite fluid, ventilatory and pharmacologic therapy, coupled with worsening ventricular dilation and contractility on echocardiography, or increasing serum lactate levels, ECMO is instituted.

Cannulation of the right atrium and aorta is performed by a cardiac surgeon via a sternotomy incision under general anesthesia in the pediatric intensive care unit. When possible, the cannulas are positioned at either end of the wound so as to allow skin closure around them to facilitate hemostasis. The rest of the chest is left open and covered with a Silastic membrane. During sternal closure, a mediastinal swab is routinely sent to the laboratory for microscopy and culture.

Constrained vortex pumps (Rotaflow, Maquet, Hirrlingen, Germany) and hollow-fiber oxygenators (Quadrox, Maquet) are used. Two primed ECMO circuits are always kept on standby for immediate use. The time from call to cannulation is generally within 15–30 mins. Currently, targeted flows are generally >150 mL/kg/min in children weighing <10 kg and

>2.4 L/min/m² in children weighing >10 kg. However, ECMO flows are individually goal-directed to targets such as normal lactate, venous oxygen saturation (>70%), mean arterial pressure, and reversal of multiorgan dysfunction syndrome. Inlet pressure monitoring (≥ -20 mm Hg) and plasma-free hemoglobin measurements are used to detect inadequate inflow, excessive pump revolutions, or hemolysis. Intravenous unfractionated heparin is administered to achieve an activated clotting time (MAX-ACT, Helena Laboratories Corporation, Beaumont, TX) of 160–180 secs. In cases of significant bleeding, the activated clotting time target is reduced to 140–160 secs, the platelet count is maintained at $100\text{--}200 \times 10^9/\text{L}$, and consideration is given to tranexamic acid administration (10–15 mg/kg q 8 hrs). Once ECMO is established, inotropes are weaned off as tolerated and ventilatory settings are minimized so as to promote cardiac and pulmonary recovery (7).

Statistics are quoted as medians and interquartile ranges, unless otherwise specified. Factors related to survival were explored, using Student's *t* test, Wilcoxon rank-sum test, or chi-square test, as appropriate. A software package (Stata 11, Statacorp LP, College Station, TX) was used to perform statistical calculations.

RESULTS

Twenty three (9.3%) patients, 13 boys and 10 girls, were included from a total of 246 ECMO runs in the 9-yr period studied. The median age and weight was 6 yrs (2.8–12.3 yrs) and 20 kg (14.5–43.5 kg), respectively. All patients had microbiological evidence of infection (Table 1). Twenty-two (96%) patients had failure of at least three organ systems, with a median number of failed organs of 4 (3–4.5). Thirteen (57%) patients required continuous renal replacement therapy. Patient

Table 2. Parameters before extracorporeal membrane oxygenation

Mean airway pressure, cm H ₂ O	16.1 (13.4–23)
Positive end-expiratory pressure, cm H ₂ O	10 (8–10)
PaO ₂ /FIO ₂	99 (50.7–192)
pH	7.11 (7.08–7.2)
Arterial lactate, mmol/L	7.8 (4.1–9.7)
Mean arterial pressure, mm Hg	57 (47–69)
Urine output, mL/kg/hr	0 (0–0.9)
Mean inotrope score	82.2 (SD, 115.6)

Statistics are given as medians (interquartile range), unless otherwise stated.

Inotrope score = dose of [dopamine + dobutamine (in $\mu\text{g/kg/min}$)] + dose of [epinephrine + norepinephrine (in $\mu\text{g/kg/min}$)] $\times 100$ (12).

circulatory and ventilator parameters before ECMO are shown in Table 2. All patients received infusions of at least two inotropes. Additionally, ten (43%) patients received adrenaline boluses, five (22%) received arginine vasopressin infusions, and one (4%) patient received a milrinone infusion. Eight (35%) patients suffered cardiac arrest and required external cardiac massage before ECMO.

Patients were placed on ECMO a median duration of 7 hrs (3–15 hrs) after admission to the pediatric intensive care unit and 9.8 hrs (5.8–26.5 hrs) after tracheal intubation. Patients received ECMO for a median duration of 93 hrs (43–119 hrs), and none required a second ECMO run. ECMO cannula sizes are shown in Table 3.

Eleven (48%) patients developed visible clots in the circuit, and two (9%) patients required circuit changes, including one patient who developed hemolysis (plasma-free hemoglobin >1 g/L). Seven (30%) patients bled from the sternotomy site and required surgical reexploration. One patient developed an upper gastrointestinal hemorrhage, which settled with conservative treatment. No patient developed mediastinitis.

Eighteen (78%) patients survived to be decannulated off ECMO. A 15-yr-old girl with staphylococcal shock was placed on central ECMO for 6 days and then converted to peripheral veno-venous ECMO for support of ongoing respiratory failure, but treatment was withdrawn after an additional 37 days of ECMO, and she died. An infant with meningococemia developed an intracranial infarction and had treatment withdrawn after 12 hrs of ECMO at the parent's request. Three other patients died of fulminant shock

Table 3. Extracorporeal membrane oxygenation cannula size and flows

Weight (kg)	Atrial Cannula (Fr)	Aortic Cannula (Fr)	Maximum Recorded Circuit Flow (L/min)
<10	14–28	10–16	1.5
10–20	24–37	16–21	4.0
21–40	28–46	18–21	4.46
41–60	28–51	20–24	7.26
>60	36–56	22–24	10.0

within 24 hrs of ECMO initiation despite maximum support. One child later died in hospital after decannulation, leaving 17 (74%) children who survived to hospital discharge. The median duration of pediatric intensive care unit and hospital stay was 233 hrs (188–377 hrs) and 549 hrs (296–1074 hrs), respectively. Higher pre-ECMO arterial lactate levels were associated with increased mortality (Table 4). No other statistically significant associations relevant to survival were demonstrated.

DISCUSSION

ECMO is not curative therapy in sepsis but rather supports the function of the heart and lungs until the causative organism can be identified and treated. The patients in our series generally required ECMO very early in the course of their rapidly progressive illnesses, usually before it could be reasonably assumed that appropriate antibiotics could have had a significant effect. In other words, ECMO was used to “buy time” for the antibiotics and other treatments to work (8).

Before the year 2000, children with septic shock requiring ECMO at our institution were put on peripheral ECMO. Preliminary data comparing the first 11 patients on central ECMO to these historical controls suggested improved survival to hospital discharge in the central group (73% vs. 38%, $p = .05$) (7). The results of this current study, with more than double the number of patients, support our initial impression that central cannulation is associated with substantially better outcomes. The theoretical benefits of central cannulation include safely achieving higher ECMO flow rates, potentially reversing shock and multiorgan dysfunction syndrome more quickly than might be accomplished by other cannulation strategies. Goal-directed therapy has been shown in other studies of sepsis to improve patient outcomes by hastening

Table 4. Characteristics of survivors versus nonsurvivors to hospital discharge

	Survivors (n = 17)	Nonsurvivors (n = 6)	<i>p</i>
Age, yrs	6.9	6.6	.908
Weight, kg	30.9	27.3	.768
Pre-ECMO mean arterial pressure, mm Hg	58	53	.539
Pre-ECMO pH	7.2	7.0	.151
Pre-ECMO arterial lactate, mmol/L	6	11.7	.007
Pre-ECMO inotrope score	60.4	144	.131
Pre-ECMO external cardiac massage, number of patients (%)	6 (35)	2 (33)	.93
Hemofiltration, number of patients (%)	9 (53)	4 (67)	.56
Organ failure, number of organs	3.8	4.2	.342
Maximum recorded flow, mL/kg/min	157.3	110.7	.055

ECMO, extracorporeal membrane oxygenation.

Inotrope score, dose of [dopamine + dobutamine (in $\mu\text{g/kg/min}$)] + dose of [epinephrine + norepinephrine (in $\mu\text{g/kg/min}$)] $\times$ 100 (12).

the resolution of shock (13), and we speculate that this is the principal reason for the relatively high survival rates. This is supported by the observation that the mean highest recorded ECMO flow rate in the first 24 hrs was higher in the children who survived. Other benefits of central ECMO may include having the return cannula in the ascending aorta, allowing higher flow rates and possibly improving coronary and cerebral oxygenation, hastening myocardial recovery and decreasing the chance of neurologic injury when compared with peripheral ECMO (14). There may also have been other factors unrelated to ECMO cannulation that contributed to the improvement in survival over time, such as better circuit technology and general improvements in critical care.

An important question is whether the patients in this study might have survived without ECMO. Although clearly impossible to answer convincingly without conducting a randomized, controlled trial of ECMO in septic shock, it seems to us very unlikely that nearly three quarters of them would have survived without ECMO. All had severe shock, were receiving high-dose inotropes, and had worsening acidosis, hypotension, and organ dysfunction despite aggressive intensive care. Clinicians caring for these children face a significant challenge. On the one hand, patients should not be placed on ECMO if they might survive without it. On the other, delaying ECMO in those who need it in order to try all other forms of less-invasive therapy increases the risks of the patient deteriorating further and suffering cardiac arrest. In our series, we seem to have erred more toward the

latter, as over a third of the patients had suffered cardiac arrest before being cannulated. Additionally, higher lactate levels were seen in those who subsequently died, perhaps reflecting relatively late initiation of ECMO in patients with more advanced shock.

Our institution is not the first to successfully employ central ECMO for sepsis (15), but it is not widely used and remains controversial. Detractors of the technique maintain that it is associated with excessive bleeding, may cause mediastinitis, and is too invasive, and that high-flow ECMO can be achieved just as adequately using two peripheral venous cannulas and a peripheral arterial cannula. In our series, although bleeding was relatively common, it was amenable to appropriate treatment and no patient died or seemed to have come to long-term harm as a consequence of hemorrhage. Although mediastinitis is clearly a risk of central ECMO, we did not diagnose it in any patient despite routine surveillance. Our institution has used high-flow peripheral cannulation in the past; although feasible in some patients, flows were not as consistently high and outcomes were not as good (7).

Our study has a number of weaknesses, including the absence of a control group, the retrospective study design, and the inclusion of only one center, which limits the strength and applicability of our conclusions. However, a prospective, randomized, controlled trial of ECMO in pediatric septic shock is not a realistic prospect (8), and we are unaware of any other center in the world that preferentially performs central ECMO in children with septic shock.

CONCLUSIONS

The survival rates of patients in this study were approximately double that of our institution's historical controls and higher than the survival rates anticipated by the American College of Critical Care Medicine. Central ECMO seems to provide better survival than conventional ECMO, and we recommend that it be considered by clinicians as a viable strategy in children with refractory septic shock.

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